

P013 Manual

RESEARCH ASSISTANT MANUAL

Study Name: Validation of a Novel Optical Sensor for Consciousness Detection via Concurrent EEG/Polysomnography in Healthy Adults

Protocol Number: 013

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Quick Links

Links	Purpose/Explanation
P013 Sentiometer Screening Form	Screening Form
https://docs.google.com/spreadsheets/d/1gWHPUWBgriFmTUxNu7S12MBaePpATTxUWRfqNInC9Qk/edit?resourcekey=&gid=108318018#gid=108318018 Can't find link	Screening Response
P013 Participant Management	Participant Management Hub
P013 Pre-Questionnaires	Pre- Questionnaires
P013 Post Questionnaire	Post- Questionnaires
https://docs.google.com/presentation/d/1mpfOXm0_-OvOsEsC0zzBsPT0Z5QvTYnhkAm2vQIWgzE/edit?slide=id.g3d6c71d6e81_0_1149#slide=id.g3d6c71d6e81_0_1149 Can't find link	Instructional Powerpoint

Subject Sign-up Flow and Scheduling

Calendly Notes:

Subject Arrival Preparations

Preparations before Session

In-Session

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A. EEG Cap Set Up

Setting up Computer

C. CGX AIM-II Set Up

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B. Sentiometer Set Up

Device Parts and Connections

Set up on Participant

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START LSL STREAM BEFORE ALL NEXT STEPS

Part 2: Pre Questionnaires and Behavioral Tasks

Part 3: NAP 🛌 ~120 mins

Sleep Preparation

Part 4: Wake Up ☀️ ~10 mins

Part 5: Post Questionnaires ~10 mins

Once the session is complete:

Subject Departure

Remuneration:

Post Session Clean Up Instructions

Plug In Equipments!!

Research Assistant General Guidance

- Practice good “bedside manner” with the subject, continually asking if they need anything while also letting them know that certain activities (e.g. using the bathroom after the EEG is on) are not preferred and could interfere with the results of the study. Be sure to let the subject know their participation is voluntary and they can withdraw at any time.
- Ensure adequate study preparation and complete data transfer after the participant leaves

Subject Sign-up Flow and Scheduling

1. Participant signs up via our landing page:
2. Participant screened and received e-consent
3. If participants are disqualified, they will get a “thank you” email
4. Participant sent a Calendly link to schedule a session time:

Calendly Notes:

- For P013, Yaya and Sam will coordinate with the EEG Assistant to block out availability/ “working hours”.
 - Use “Assigned RA” to indicate the available/responsible days
 - Use “Unavailable RAs” to prevent a timeframe from being scheduled if it is your assigned days, and there are other events going on (meetings, etc)
 - Other scheduled lab events will automatically be checked for conflicts on Calendly.
 - Calendly Workflow
 - Subject signs up for event → Calendly Event Confirmation Page → Event blocked off on IACS Experiments Calendar → New event email sent to RA gmail account → Invitee gets a calendar invite from ra@advancedconsciousness.org Google Calendar
 - (Workflow) 3 days before event - Subject gets a confirmation email → ask to reconfirm
 - (Workflow) 2 days before event - Subject gets a confirmation email + reminder about nap preparations
 - (Workflow) 24 hours before the event - Subject gets a confirmation text + reminder about nap preparations
 - RA Responsibility: send participant a personal reminder text a few hours before session starts
 - When a participant cancels, it will automatically trigger Calendly cancellation and scheduling link emails. We will get email and slack notification of cancellation
 - If RA is cancelling or rescheduling a session, **do it through Calendly**.
 - No shows
 - a. If a subject cancels shortly before their appointment or does not confirm appointment → Cancel appointment in Calendly or Calendar
 - b. If a subject confirms but no shows at their appointment time → Mark as no show in management hub and reach out to participant to schedule a new time
-

Subject Arrival Preparations

1. 12 hours before the event → participant gets a text reconfirmation
 - a. Note that this is not reflecting on the Gcal invitation - we will get a reconfirmation notification in the RA Gmail if they reconfirm
2. Before the session, check the RA email and see if there is a reconfirmation
 - a. If yes, then all good
 - b. If not, then the participant did not reconfirm that we need to text them to double check

3. RA: When getting to the lab, please send a personalized text 1-2 hours before start time

“Hi [subject's name], this is [RA's name] from the Institute for Advanced Consciousness Studies. I will be with you for your study session today. If you have trouble finding the parking or suite 510, reach out to [this phone number].”

Preparations before Session

- Double check to ensure that the EEG is plugged in
 - Prep CGX skintacts
 - Make sure the bed is set up properly, Oslo charged and eye mask prepped
 - Open the instructional PPT on Mac
 - Set up Mac to be ready for participant to operate
 - Prepare all set up equipments: experiment computers, EEG cart, CGX Aim-II, Sentiometer
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In-Session

Part 1: Set Up ~45 min

Note: during EEG set up, participant will be viewing the instructional PPT

A. EEG Cap Set Up

Preparing Subject

Location: office chair, facing Mac

1. Have the subject remove any earrings/necklaces/jewelry/watches and place at **Subject Desk 1** for safe keeping.
2. Briefly describe to the subject what's next in terms of Neuroprep, electrode gel and EEG cap setup
 - a. RA Script: “We’re going to measure your head to get an EEG cap size. We will use cleansing gel to remove oils from the scalp and then apply electrode gel in each of the channels. If you have sensitive or dry skin, you might feel some discomfort during the scrubbing process, but you should not feel any burning. Let us know if you feel pain at any point.”
3. Direct subject to sit in office chair, placed in front of the recliner.
4. Measure the subject’s head from the nasion to the inion (notch between nose and forehead to the bony protrusion on back of the scalp) using the tape measure. 10% of that distance is the expected distance from the nasion to the most anterior channel FPz.

a. On the tape measure, the solid lines indicate a centimeter mark.

5. Choose the cap / Electrode Input box combination that would best fit their head by measuring fully around the skull from the eyebrow line to the inion. Err on the side of rounding down so that the cap will be tighter instead of looser.

a. You can see the size of each cap on the label.

6. Put on gloves before touching subject.

7. Ask subject to briefly remove their mask while you put the EEG cap on, they will put it back on momentarily (after cap is velcroed)

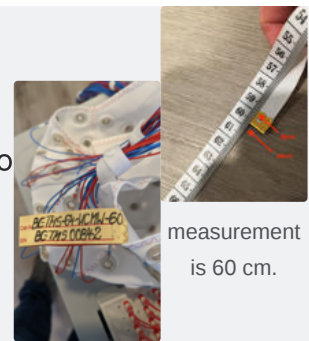
8. Place the cap on subject's head.

a. Begin at the forehead, ask subject to help by holding their finger to their forehead to keep cap in place as you place it over their head.

b. Make sure the cap is flat everywhere on the scalp. Readjust hair if need be and make sure ears are coming through the ear holes.

9. Hand chinstraps to subject and ask them to use the Velcro to secure the strap under their chin. They can then replace their mask.

a. RA Script: "Pull the straps firmly down and forward, then attach the Velcro under your chin as tightly as possible without being uncomfortable. You can then put your mask back on."

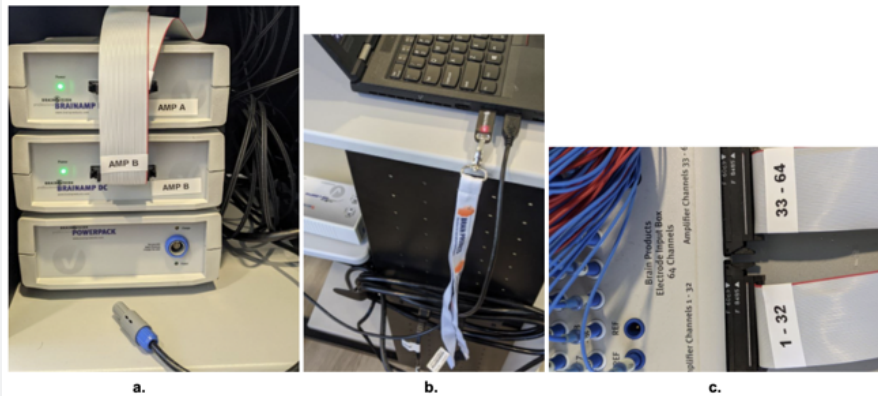


this is a 60cm
cap

Setting up Computer

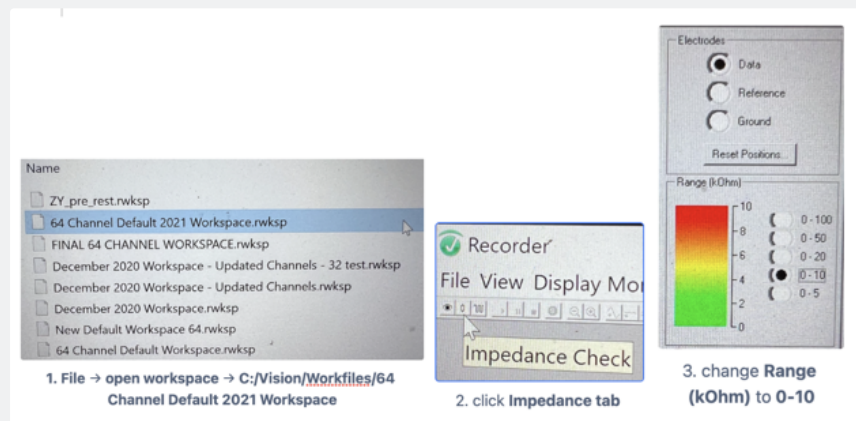
Location: EEG Cart and Computer 3. Set up EEG with cart parallel to the subject.

1. Ensure all connections are appropriately wired
 - a. Make sure amplifiers are unplugged from the battery and green light is on.
 - i. Double check that the ribbon cable for Amp A is connected to Amp A and same for Amp B
 - b. Make sure Dongle (with white lanyard) and USB 2 Adapter are both plugged in to the USB ports.
 - c. Make sure the ribbon cables are connected to the correct Amplifier Channels



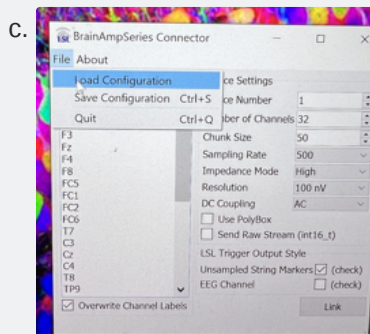
ensure all connections are appropriately wired

2. Cycle through each electrode hole and try to move hair aside and/or out of the hole so that you can see the scalp.
3. Open Recorder (green Checkmark logo)
 - a. “Do you want this app to make changes to your device?” prompt will appear for Recorder.exe → yes
 - b. **File > Open workspace > Desktop > Config and workspace files > for brainamp recorder > 64 channel default 2021 workspace**



4. Get all impedance values to turn green (0-10kOhm range)
 - a. **Start with the Ground and Reference electrodes first.** They need to turn green before the other electrodes have the chance to Nuprep and electrode gel.
 - i. Dip cotton swab in Nuprep. Cycle through each electrode hole and try to move hair aside and/or out of the hole so that you can see the scalp.
 - ii. Insert the syringe into the channel you’ve created to the scalp and build a column of gel connecting scalp to electrode
 - b. The true trick to a successful EEG placement is to:
 - i. Make sure the cap is flat everywhere on the scalp. Readjust hair if need be and make sure ears are coming through the ear holes.
 - ii. Use the wood part of the Qtip to part the hair so that you can see the scalp

- iii. Put a small bit of Nuprep on the tip of the Qtip and apply directly to scalp, then give a few twirls back and forth and clockwise / counterclockwise. The point here of the Nuprep is to help remove oil and dead skin from the scalp, which dramatically lowers impedance levels.
 - iv. Repeat #3 for all electrodes. This is easily the most time intensive part. **The NuPrep is crucial.**
 - v. Apply gel to the spot where you can see the scalp and let it flow up through the hole until you see it at the top. Err on the side of conservation with the electrode gel. Then press down on the electrode with the gel in it into the head, gently.
 - vi. Repeat #5 for all electrodes. You should be able to do this very quickly and get all greens!
 - vii. Target problematic electrodes by making sure to repeat steps 2, 3, and then 5.
 - c. If, after applying gel to each electrode, you do not have green impedances across all channels, feel free to move on to the Bioperipherals setup. Sometimes, the impedances will decrease as a function of time because the electrodes will start to settle into the hair.
 - d. If after the bioperipherals are set up, you do not see all green impedances, try filling with more syringes and swirling the area under the electrode to get good connection.
 - e. If all fails, you can use the Nike swim cap to wrap the EEG cap, tightening it against their scalp. This should create better contact and lower impedances.
 - f. **Aim for all green 0-10.** If that's not possible, all green 1-100. No yellows and definitely no reds.
5. Once all impedance values are green, press Stop Monitoring (red stop sign button) in order to close BrainVision Recorder.
 6. EEG Computer: Start Brain Amp Series Connector to create the LSL Stream (Grey BrainAmpSeries App)
 - a. **File → Load Configuration: Select Desktop > Config and workspace files > for brainamp connector > Brain Amp Series Connector Configuration File -**
 - b. If you press “Link” without going to File → Load Configuration, then “Number of Channels” box will show 32 (and only 32 channels of data will be collected)



d. **Ensure: “Number of Channels” box: 64 (not 32)**

e. Press “Link” box at the bottom once this is done

Note: expected LSL stream is called BrainAmpSeries-Dev_1

C. CGX AIM-II Set Up

- Ensure CGX Dongle is connected to **Computer 1**
- Connect sensors to subject’s body
 - a. Clean the skin with an alcohol wipe before placing the SkinTacts.
 - b. Be sure to explain what and where you will clean with the alcohol wipe before touching participant
 - i. Narrate what you are doing at **each** body part: “Now I am going to wipe your hand,” “Now I’ll place the sensors on the back of your neck,” etc.
 - ii. Exercise extra care with respiration sensors under the collarbone (follow the angle of the collarbone)
 - c. Peel off SkinTacts just before attaching to the subject.

List of Channels and Placements

1. **Ground [GND]**

- This should be done first.
- Place on left side of the neck

2. **Bio-Impedance Based Respiration [RESP+ / RESP-]**

- Attach paddles just below the collarbone of the subject
- Follow the angle of the collarbones
- Make it such that the arrows face each other (towards the midline). There is only one placement orientation for the arrows to face each other.

3. **Eye [EXG 1+ / EXG 1 -] and [EXG 2+ / EXG 2-]**

- Attach  paddles by the eyes (near the temple)
- Attach the  paddles behind the ear at bony bumps

4. **Chin [EXG 3 + / EXG 3 -] and [EXG 4 + / EXG 4 -]**

- Use nuprep after alcohol wipe
- Use tape and gaze to secure the skintacts

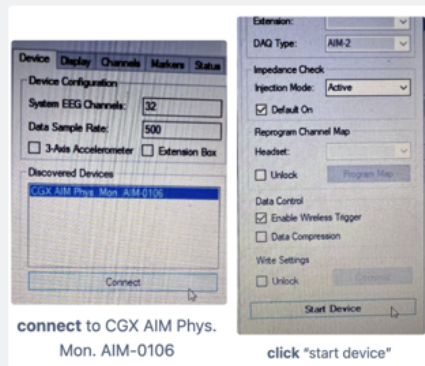
- Attach + paddles by on the chin, one on either side of the midline of chin
- Attach the - paddles under the chin, parallel to its positive counterpart



Left to Right: GND + RESP, Eye, Chin

CGX Quality Check

- **Computer 1:** Launch CGX Acquisition
- Turn on CGX Device
- Connect to CGX AIM Phys. Mon. AIM-0106 in CGX Acquisition Software → Start Device



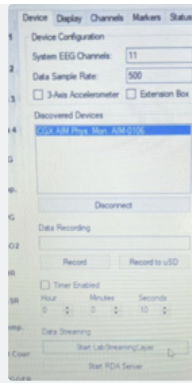
Perform Quality Assurance

1. Bio-Impedance Based Respiration
 - a. CGX Acquisition → Display → Channels (2) → Show Page (EEG 5 to 6)



2. Eye and Chin

Start CGX LabStreamingLayer



Note: expected LSL stream is named CGX AIM Phys. Mon. AIM-0106

B. Sentiometer Set Up

Device Parts and Connections

1. Sensor
2. Console
3. USB cable
4. VGA Cable
5. Recording System-Laptop
6. Velcro Strap
7. Tape

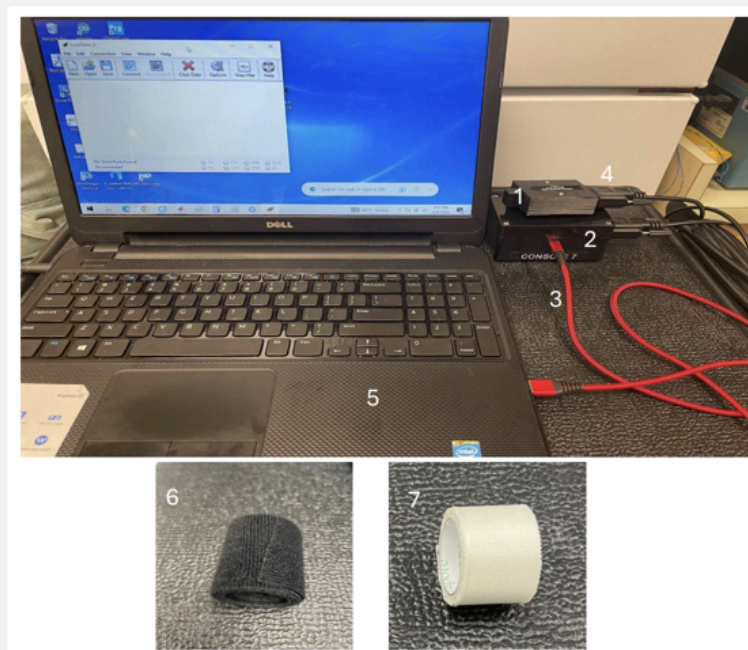


Figure 1.1: Sentiometer Parts, their connections and accessories.

Set up on Participant

1. The top of the sensor cover features the product label, while the bottom is equipped with a Velcro attachment strip.
2. Secure the Velcro strap to the bottom of the sensor, ensuring it matches the orientation in Figure 2.1a

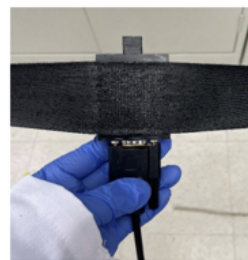


Fig 2.1a: Bottom view of sensor attached to Velcro Strap



Fig 2.1b: Top view of sensor attached to Velcro Strap

3. Secure the sensor to the volunteer's non-dominant upper arm, midway between the shoulder and elbow, with the 'BD sensor' label facing the skin. Ensure the soft side of the Velcro strap is against the arm for comfort. Wire facing down.



C) Recording System Setup

Step 1: NOTE PLUGGING POSITIONS

Step 2: **Task Computer 1: Sentiometer-P013 folder → sentiometer-stream (windows batch file) → IACS Sentiometer – Session Setup pops up in terminal**

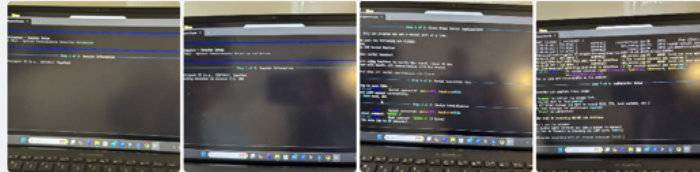
- This set up terminal basically walks you through the Sentiometer set up process

Step 2: Enter participant ID (13##) and recording duration (300 mins)

Step 3: Confirm that the Sentiometer is showing up as COM4

Step 4: Follow instructions in Sentiometer stream, using enter and y to confirm

- Is COM4 the Sentiometer [Y/n]: y
- Does the data look reasonable [Y/n]: y
 - Once the Sentiometer starts previewing and showing the samples, the rightmost Hz column should be in the 500 range



Reference here:

Note: confirm that the stream showed up in LSL as Sentiometer (LAPTOP-7LJUD7F8)

⚠ START LSL STREAM BEFORE ALL NEXT STEPS

Streams Checklist:

☐ Ensure there are 6 total streams in Lab Recorder

1. BrainAmpSeries-Dev_1-Markers
2. BrainAmpSeries-Dev_1
3. CGX AIM Phys. Mon. AIM-0106
4. CGX AIM Phys. Mon. AIM-0106 Impedance
5. Sentiometer (LAPTOP-7LJUD7F8)
6. P013_Task_Markers (won't show up until P013 Session Launcher is started)

Make sure the intercom is set up before leaving the room!

- Intercom on Channel 1
- Hold down “talk” to communicate: red light comes on = can start speaking
 - Wait until red light comes on as you're holding down “talk” otherwise the beginning part of your speech might not go through

Note: RA will not be in the room during questionnaires and behavioral tasks. Make sure participant are very clear with the instructions before you leave the room.

→ Instruct participant to alert us when they are done with the behavioral tasks on the intercom by holding down “talk”

During this hour:

- Set timer to keep track of portion time
- Attend to intercom in case participants have questions
- Set up silent doorbell on front door
- Change sign on front door to be “meditation study in progress”
- Set up indication lights for SILENCE
- Send message in Slack channel to alert people who are in lab

Part 2: Pre Questionnaires and Behavioral Tasks

- Seat participant at study desk in front of iMac
- The questionnaires and behavioral tasks are both combined into one python script. After questionnaires are complete, the script moves straight into the behavioral battery
 - Trait (FFMQ, PSQI, ESS) and State (SSS, POMS) Questionnaires
 - Behavioral task battery: Auditory oddball, RGB Illuminance, Perceptual Awareness Threshold, Mind-state switching, SSVEP

• To launch:

- GITHUB (folder on desktop) → iacs-sentiometer-study → Launch P013 (Application)

• P013 Session Launcher (GUI) pops up after launching

- Takes some time for the GUI to pop up in center of the screen
- Enter Participant ID
- Pre-Launch Checklist
 - Note: If Vayl isn’t showing “OK” then the Vayl application needs to be opened. It is usually already opened/running
- Press Start Session button

→ Set timer on phone for approximately 40-45 minutes to know when the participant will be done with the behavioral tasks. They will alert us via intercom when they are finished

Part 3: NAP 🛏️ ~120 mins

Sleep Preparation

- Transition to bed
- Ozlo Sleepbuds insertion and eye mask as needed
 - Set up Ozlo sleepbuds with alarm set for 2 hours and white noise if the participant wants
 - Use Ozlo app on purple iPad and follow app instructions

- Lights off

→ **Set 2 hour timer on phone/computer to know when to enter the room again and wake up participant from nap**

Part 4: Wake Up ☀️ ~10 mins

- Wake-up (Ozlo Sleepbuds alarm); timer
- Breakdown
- Bathroom, Refresh
- EEG Removal
- CGX skintacts removal
- Stop Lab Recorder stream immediately
- Stop Sentiometer stream

Part 5: Post Questionnaires ~10 mins

- Seat participant at study desk in front of iMac
- Administer “P013 Post-Questionnaire” bookmarked on Chrome
 - (SSS, POMS), Sleep Quality VAS, Brief Dream Report
- Let them know their Participant ID
- Be available for any questions participants may have

Once the session is complete:

1. End the timer that indicates how long the session took
2. Note this on the management google sheet

Subject Departure

- Thank the participant for their time
- Parking validation tickets are given out after each session
- Reimbursement for the study (\$30/hour) is paid in total

Remuneration:

- Compensation will be provided via cash or Venmo (participant preference) at the conclusion of the session. Parking validation tickets are provided separately and are not cash-equivalent.

- Participants who withdraw before completing the session will receive prorated compensation for time spent, calculated at \$30/hour rounded up to the nearest 15 minutes.

Item	Value
Session duration	~4.5 hours
Hourly rate	\$30/hour (prorated to 15-minute intervals)
Total compensation	\$135.00
Parking validation	\$31.50 value (provided via physical validation tickets)
Total value per participant	\$166.50

Post Session Clean Up Instructions

- Stripping the bed (everything aside from the comforter)
- Throw into the laundry basket
- Cleaning the space with Lysol wipes
- Wash and clean the EEG cap
- Clean Ozlo sleepbuds

Plug In Equipments!!

→ This is very important at the end of the session. All the equipment needs to be at full charge for each study session

Make sure all of the power cables are plugged in (EEG cart, EEG power bank, CGX device, and charging cables for the computers) upon completion of an experiment.

EEG

1. Make sure the amplifiers are plugged in to the battery and the battery is plugged into the wall. The green light on each amplifier should turn off. This step is super important. If the amplifiers are not charged, we cannot conduct the experiment.



2. Wash the EEG cap at the kitchen sink, keeping the base (where all the red and blue wires are) as far away from the sink as possible.



DO NOT GET WATER on the base →
cover with a towel

- ☐ Plug in CGX-AIM II
- ☐ Make sure all 3 computers are charging – EEG computer, Task Computer, Lab Recorder computer
- ☐ Plug in Ozlo Sleepbuds
- ☐ Plug in light towers that are placed around the lab
- ☐ Organize Sentiometer (and wires) and place it back in the black hard case